Nanhao Su

RESEARCH PORTFOLIO | Trustworthy AI | Agent Memory

An undergraduate researcher interested in how LLM-based agents maintain and use information reliably, and how their behavior can be evaluated with traceable evidence.

Huazhong University of Science and Technology | Wuhan, China | Expected graduation: Jun 2029

RESEARCH INTERESTS

Agent reliability, safety & evaluation

I work with a trustworthy AI research group at HUST and explore long-term agent memory, AI system security, and controlled evaluation of model-driven behavior. I prioritize reproducible experiments, source-level evidence, and clear separation between observed outcomes and untested hypotheses.

CURRENT RESEARCH

HUST Trustworthy AI Research Group | Undergraduate Researcher | May 2026 - Present

Agent memory reliability & empirical evaluation

- Investigated A-MEM and a fixed-version Mem0 implementation through controlled synthetic-data experiments, focusing on long-term memory updates, state consistency, and retrieval behavior.
- Documented 54 A-MEM test configurations and 72 matched replay units; maintained auditable traces and preserved unsuccessful cases.
- Developed preliminary evaluation protocols and pilot studies related to long-horizon reliability in memory-enabled agents.

Provincial Undergraduate Innovation Project | Research Team Member | 2026 - Present

Telecom-fraud dialogue augmentation

- Adapted data-augmentation approaches from TeleAntiFraud-28k (ACM MM 2025) and Bot Wars Evolved (2025).
- Used Qwen and DeepSeek on a pre-existing corpus of 66,699 fraud case descriptions; contributed to generating 10,000 annotated synthetic dialogues.
- Contributed to prompt refinement, data-quality checks, and fraud-detection evaluation workflows.

Selected public projects

RESEARCH PORTFOLIO | Public engineering artifacts

AGENT RISK AUDIT

Reproducible offline tooling to audit structured agent risk-rating records, preserve provenance, and distinguish missing, duplicate, and invalid entries.

11 unit tests | 4 blocked probes | Synthetic fixture

github.com/nanhaosu/agent-risk-audit

FLOWOPS ENGINEERING

A synthetic incident-coordination prototype with explicit execution constraints, persistent sessions, and recovery obligations. Designed to make state changes and negative results inspectable.

45 Python tests | 9 Node tests | 36 synthetic trajectories

github.com/nanhaosu/flowops-engineering

METHODS & SKILLS

Research practice

Controlled comparisons, synthetic test scenarios, trace audits, matched replay, negative-case documentation, data quality assessment.

Technical toolkit

Python, C, PyTorch, Hugging Face Transformers, Git, Linux, Selenium, SQLite, LaTeX.

AVAILABILITY

Seeking remote research collaboration from January 2027 onward. Interested in empirical AI safety, agent memory, and reproducible evaluation, with sustained academic mentorship.

CONTACT AND LINKS

University email: sunanhao@hust.edu.cn

GitHub: github.com/nanhaosu